1) $R = N \times f$

$= 1024 \times 50Hz$

$= \boxed{51200 \text{ spikes/sec}}$

2) $B = R \times 20$

$= 51200 \times 20$

$= 1,024,000$ bit/sec

$= \boxed{1.024 \text{ Mbit/sec}}$

	Rate	Y/N
SPI	≤ 50	Y
I_2C	≤ 3.4	Y
AXI4-L	≤ 100	Y

3) For SPI $\leq$ 50Mbit/s : Yes it can sustain the mean rate of 1.024Mbit/s

For $I_2C \leq$ 3.4Mbit/s : Yes it can sustain the mean rate and this is the chosen lowest complexity interface.

For AXI4-Lite ~ 100Mbit/s : Yes it can sustain the mean rate of 1.024Mbit/s

4) $1024 \times 0.25 = 256$ spikes

$256 \times 20 = 5120$ bits in window

$5120 / 0.001 = 5,120,000$ bit/sec

$= \boxed{5.12 \text{ Mbit/s burst-peak bandwidth}}$

$5.12 / 1.024 = \boxed{5 : \text{burst-to-mean ratio}}$

Compared to the chosen interface in part 3 of $I_2C \leq$ 3.4 Mbit/s it would need a buffer since the 5.12 Mbit/s burst exceeds the 3.4Mbit/s the I_2C can handle

(5) $1024/0.001 = 1{,}024{,}000$ bit/s
$= \boxed{1.024 \text{ Mbit/s frame-based bandwidth}}$

$1.024/1.024 = \boxed{1 = \text{Ratio AER/frame}}$

$N \times f \times 20 = N \times 1000$
$f \times 20 = 1 \times 1000$

$f = 1000/20$
$f = \boxed{50\,\text{Hz crossover firing rate}}$

When AER is the right choice when the neural network is highly sparse and the average firing rate is below the 50Hz crossover threshold.